Answers: The chain rule

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Summary

Answers to questions relating to the guide on the chain rule.

These are the answers to Questions: The chain rule.

Please attempt the questions before reading these answers!

1.1. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\dfrac{1}{7}\cos(5+4x)\right) = -\dfrac{4}{7}\sin(5+4x).$

1.2. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(4\cos(x^2)\right) = -8x\sin(x^2).$

1.3. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(e^{x^2+5}\right) = 2xe^{x^2+5}$

1.4. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(2(\sin(2x))^2\right) = 8\sin(2x)\cos(2x).$

1.5. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(e^{\sin(3x)}\right) = 3\cos(3x)e^{\sin(3x)}.$

1.6. Using laws of logarithms, write $\ln((2+4x^{-2})^{-1}) = -\ln(2+4x^{-2})$. Then

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln(2+4x^{-2})^{-1}\right) = \frac{8x^{-3}}{2+4x^{-2}} = \frac{4}{x^3+2x}.$$

1.7. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(e^{5x^4}\right) = 20x^3e^{5x^4}$

1.8. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(e^{2x^{-3}}\right) = -6x^{-4}e^{2x^{-3}}$

1.9. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(-5\sqrt{x-2}\right) = -\dfrac{5}{2\sqrt{x-2}}$

1.10. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{(x+3)^2}\right) = 1$ (for $x \geq 3$.)

1.11. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\ln(x^2+1)\right) = \dfrac{2x}{x^2+1}$

1.12. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\ln(\cos(x))\right) = -\tan(x)$.

1.13. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(2\cos^2(x)\right) = -4\cos(x)\sin(x) = -2\sin(2x)$.

1.14. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(2(x^3+5x^2+13x-1)^3\right) = 6(3x^2+10x+13)(x^3+5x^2+13x-1)^2$.

1.15. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{\dfrac{1}{2x}}\right) = -\dfrac{1}{4x^2}\left(\dfrac{1}{2x}\right)^{-1/2}$

1.16. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\cos(5x^{-1/2})\right) = \dfrac{5}{2}x^{-3/2}\sin(5x^{-1/2})$.

1.17. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\sin\left(\sqrt{x^2+1}\right)\right) = \dfrac{x\cos(\sqrt{x^2+1})}{\sqrt{x^2+1}}$

1.18. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\sin(e^x)\right) = e^x\cos(e^x).$

1.19. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\cos(e^{-2x}+5)\right) = 2e^{-2x}\sin(e^{-2x}+5).$

1.20. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\ln(3x^3+\sin(x))\right) = \dfrac{9x^2+\cos(x)}{3x^3+\sin(x)}.$

Version history and licensing

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